Literature Status: Helson and Embedding Problems for Dirichlet Series

Run `fa_banach_001`, lane 8

2026-07-19

Status

Status: literature_already_answered for Helson's problem, and **literature_already_answered (partial subrange)** for the embedding problem. This packet records already known answers, not new proofs from the run.

Source Question

The source paper is Eero Saksman and Kristian Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616. In Section 3, Problem `helson` asks:

$$\left\| \sum_{n=1}^{N} n^{-s} \right\|_{\mathcal{H}^1} = o(\sqrt{N}) \qquad (N \to \infty)?$$

The surrounding text explains the probabilistic interpretation: by viewing p_j^{-s} as independent Steinhaus variables, this is Helson's conjecture for sums of a Steinhaus random multiplicative function.

The same paper also asks the embedding problem in Section 2, Problem `embedding`: whether the local L^p integral of a Dirichlet polynomial on a fixed segment of the line $\operatorname{Re} s = 1/2$ is bounded by a universal constant times its $\mathcal{H}^p$-norm to the p-th power.

Supporting Answer

The supporting paper is Adam J. Harper, *Moments of random multiplicative functions, I: Low moments, better than squareroot cancellation, and critical multiplicative chaos*, arXiv:1703.06654, published in *Forum of Mathematics, Pi* 8 (2020), e1.

Harper's Theorem 1 states that for a Steinhaus random multiplicative function f, uniformly for $0 \le q \le 1$,

$$\mathbb{E} \left| \sum_{n \le x} f(n) \right|^{2q} \asymp \left(\frac{x}{1 + (1-q)\sqrt{\log\log x}} \right)^{q}.$$

Taking $q = 1/2$ gives

$$\mathbb{E} \left| \sum_{n \le x} f(n) \right| \asymp \frac{\sqrt{x}}{(\log\log x)^{1/4}} = o(\sqrt{x}).$$

Harper explicitly notes after Theorem 2 that this proves Helson's conjecture. Through the Bohr correspondence used in the source paper, this is exactly the affirmative answer to Problem `helson`.

Secondary Embedding Consequence

Harper also explicitly states that Theorem 1 gives a negative answer to the embedding problem for all exponents $0 < 2q < 2$, i.e. $0 < p < 2$. The counterexample sequence is the family

$$P_x(s) = \sum_{n \leq x} n^{-s}.$$

For fixed $0 < q < 1$, Riemann–Stieltjes integration gives

$$\int_0^1 \left| \sum_{n \leq x} n^{-1/2-it} \right|^{2q} dt \asymp x^q,$$

whereas Theorem 1 gives

$$\|P_x\|_{\mathcal{H}^{2q}}^{2q} = \mathbb{E} \left| \sum_{n \leq x} f(n) \right|^{2q} = o(x^q).$$

Thus no universal embedding constant can exist for $0 < p < 2$.

Scope

This gives a full literature answer to Problem `helson`. It gives a negative literature answer to Problem `embedding` only in the subrange $0 < p < 2$. The source paper already records the positive $p = 2$ and even integer cases; this packet does not classify the remaining exponents or the other survey problems.

Evidence

The source PDF is included as `source_paper.pdf`. The supporting Harper PDF is included as `supporting_paper_1703.06654.pdf`. The identification was checked against the source LaTeX lines 174–181 and 265–267, and Harper's Theorem 1 statement plus following explanatory paragraphs at lines 148–162 of the parsed source.

References

[1] E. Saksman and K. Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, 2016.

[2] A. J. Harper, *Moments of random multiplicative functions, I: Low moments, better than square-root cancellation, and critical multiplicative chaos*, Forum of Mathematics, Pi 8 (2020), e1; arXiv:1703.06654.